

# DehatiAI — Executive Cost Audit & Infrastructure Report

Verified Against Codebase: backend/package.json & backend/routes/ai.js

100% CODE VERIFIED

CURRENT MONTHLY SPEND

\$2.50

Prototype & Testing Phase

SCALE MONTHLY (500 DAU)

\$170.50

~\$2,046 / year total

PRODUCTION (5,000 DAU)

\$1,191.00

~\$14,292 / year total

 Master Cost & Infrastructure Audit Table (Codebase Verified)

| SERVICE              | PURPOSE                              | PRICING MODEL            | FREE LIMIT          | CURRENT | 500 DAU  | 5K DAU     | 5K DAU (YR) |
|----------------------|--------------------------------------|--------------------------|---------------------|---------|----------|------------|-------------|
| Anthropic Claude API | AI Chat, Vision, Disease, Fertilizer | Per-Token Usage          | \$0 (Pay-as-you-go) | \$2.50  | \$157.50 | \$1,125.00 | \$13,500.00 |
| Railway App Hosting  | Node.js Express API Server           | Compute (CPU/RAM)        | \$5 Trial Credit    | \$0.00  | \$12.00  | \$45.00    | \$540.00    |
| Railway PostgreSQL   | Primary Database & L2 Cache          | Storage & IOPS           | Included in Railway | \$0.00  | Included | Included   | Included    |
| Vercel               | Frontend PWA Hosting                 | Bandwidth & Builds       | 100 GB / month      | \$0.00  | \$0.00   | \$20.00    | \$240.00    |
| Supabase             | Secondary DB & Auth Backup           | MAU & Database Size      | 50,000 MAU / 500MB  | \$0.00  | \$0.00   | \$0.00     | \$0.00      |
| Open-Meteo API       | Weather Forecast Intelligence        | Free Public API          | 10,000 calls / day  | \$0.00  | \$0.00   | \$0.00     | \$0.00      |
| AMIS Mandi Data      | Commodity Market Prices              | Internal Base + Admin DB | Unlimited           | \$0.00  | \$0.00   | \$0.00     | \$0.00      |

| SERVICE           | PURPOSE                 | PRICING MODEL       | FREE LIMIT | CURRENT | 500 DAU  | 5K DAU     | 5K DAU (YR) |
|-------------------|-------------------------|---------------------|------------|---------|----------|------------|-------------|
| Custom .PK Domain | Branded Domain & SSL    | Flat Annual Renewal | N/A        | \$0.00  | \$1.00   | \$1.00     | \$12.00     |
| TOTAL BUDGET      | Complete Infrastructure | Combined            | —          | \$2.50  | \$170.50 | \$1,191.00 | \$14,292.00 |

⚖️ LLM Provider Cost Comparison (Claude 3.5 Sonnet vs Gemini 2.0 Flash)

| LLM Provider & Model                       | Input / 1M | Output / 1M | 5K DAU Monthly  | Annual Total     | Cost Impact                      |
|--------------------------------------------|------------|-------------|-----------------|------------------|----------------------------------|
| Anthropic Claude 3.5 Sonnet (Current Code) | \$3.00     | \$15.00     | \$1,125.00 / mo | \$13,500.00 / yr | Baseline (High Precision Vision) |
| Google Gemini 2.0 Flash (Alternative)      | \$0.10     | \$0.40      | \$42.00 / mo    | \$504.00 / yr    | 96.2% Cost Reduction             |
| Anthropic Claude 3.5 Haiku (Hybrid Option) | \$1.00     | \$5.00      | \$375.00 / mo   | \$4,500.00 / yr  | 66.7% Cost Reduction             |

🛡️ Architectural Safeguards & Risk Mitigation

⚡ 90% AI Token Cost Savings via Ephemeral Prompt Caching

Anthropic system prompts are cached for 5 minutes in backend/routes/ai.js (\$0.30/M cached tokens vs \$3.00/M full price). In addition, identical farmer questions are served directly from the PostgreSQL L2 database cache in ~2ms at **\$0.00 API cost**.

⚠️ Rate Limiting Traffic Protection

To protect against token drain from malicious traffic or bot loops, our backend enforces a strict rate limit of **30 AI calls per hour per IP** via middleware/rateLimit.js.